[Viettel AI / Internship] Research report
Time: December 2024

Summary

In November 2024, we learned about Endoscopy and Transfer Learning. We have created 2 models: An image quality classification and a disease classification for endoscopy video/images. We utilize Transfer Learning by using the pre-trained model from hugging face

The project showed that: - The combination of ResNet50 and DINOv1 gives the best result out of all combinations/models.
- The main problem while training the model is the Imbalance dataset and overfitting but can be resolved by under-sampling and data augmentation.
- Transfer Learning is a strong tool to tackle problems in the endoscopy field.

Introduction

This document outlines the findings from a research project conducted by Hồ Minh Quang in November 2024 for the purpose of image classification in endoscopy videos.

Methodology

Image quality classification

The method used in the research is using pre-trained models from hugging face to perform a transfer learning to classify image quality in endoscopy videos (dark, mucus, blurry , overexposure and good) .

Data will be taken from the Gastrovision dataset and will be hand-picked for the training data of the model. Not only that we will be extracting frames from this video and hand-pick frame that are relevant and give out good information for the model to train on.

We preprocessed the data by:
- All images were resized to a fixed dimension of 224x224 pixels.
- Cropped to the center of the images due to the endoscopy video have a fish eyes effect
- Data augmentation techniques were employed to artificially increase the dataset size and improve the model's robustness:

- Random horizontal and vertical flips, rotations of up to 20 degrees
- Transformations such as translations and scaling.
- Normalized using the mean and standard deviation of the ImageNet dataset to ensure consistent input ranges for the model.
- Split data to training and testing (80 - 20)

We then freeze the parameters of the pre-trained ResNet50 model. Only the parameters of the final fully connected layer are trained

It is important to note that there are a few problems that we came across such as an Imbalance dataset and overfitting due to the nature of endoscopy videos, but we have managed to deal with it.

Disease classification/position

We will use the data from the Gastrovision dataset, which has 27 classes of the disease/position inside the human body. In each class, we took around 30 images per class for the model's training data. this is because if we take the whole dataset, there will be an imbalance dataset problem due to some of the classes having so many more images than the others class.
The data Preprocessing step will be the same as the previous one

Combine 2 models

We take 2 of our best models, one for image quality classification and one for disease classification, and put them together:

- Input an endoscopy video, extract, remove similar frames, and transform the remaining frames to feed into the model. Each video will be created in a folder in Google Drive to store further image predictions.
- Processed frames will be fed to the image quality classification model to classify them into 5 classes: Blurry, dark, mucus, overexposed, and good. Each class will be assigned to a folder on Google Drive and after the model finishes classifying, frames will be saved to the corresponding folder
- We then only take the frames from the good frame folder and feed them to the disease classification model
- The model then will classify them and name the images the prediction of what these images are, then will be saved into a folder.

Related parameters

Result

Using ResNet50 architecture and pre-trained model from hugging face. We have trained and tested all given model

ResNet50 on Pretrained with DINOv1 on Gastronet-200K: 84% accuracy
ResNet50 on Initialized with Billion-Scale weights and Pretrained with DINOv1 on Gastronet-5M: 82% accuracy
ResNet50 on Pretrained with DINOv1 on Gastronet-5M: 80%
ResNet50 on Pretrained with MOCOv2 on Gastronet-5M: 41% accuracy
ResNet50 on Pretrained with SIMCLRv2 on Gastronet-5M: 41% accuracy
VIT-small on Gastronet-5M: 47% accuracy

The performance of combining 2 models is around 80% and can be improved by adding more data after each test with new videos. The reason why the model can only reach 80% accuracy is that in endoscopy videos, there are so many similarities between each frame or each disease, which can easily confuse the model and make it predict falsely

Conclusion

Overall, we have successfully classified the image quality from an endoscopy video and used the good images to detect potential disease at an acceptable performance. Our future work wil be try to increase the size of the dataset without making it imbalance . We will also implement image segmentation and then combine all 3 models.